

PERCEPTION MODELS FOR AUTONOMOUS DRIVING IN VIRTUAL SIMULATIONS



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The rapid advancement of autonomous driving systems (ADS) heavily relies on accurate perception and fusion of multi-sensor data, such as cameras, radars, and lidars. This project aims to implement a developed perception uncertainty model into virtual simulations for safety assessment of ADS.

Observe: THE PROJECT WOULD ONLY BE OFFERED IF WE HAVE SEVERAL APPLICANTS MATCHING THE PREREQUISITES. PLEASE CONTACT AMIN ASSADI OR JORDANKA KOVACEVA, IF YOU HAVE THESE QUALIFICATIONS AND ARE INTERESTED IN PARTICIPATING IN THIS PROJECT.

PROJECT OBJECTIVES

To enable effective safety benefit assessment, the project will focus on the following key objectives:

- Implement the proposed perception models in virtual assessment of ADS.
- Estimate the safety performance of ADS with ideal condition and perception uncertainty model.

LEARNING OUTCOMES:

By the end of the project, students will be able to:

1. Understand multi-modal sensor data characteristics and challenges in autonomous driving.
2. Perform data preprocessing and exploratory analysis of effect of the perception model on different test scenarios with synthetic crash data.
3. Design test scenarios with open-source standards ASAM.

4. Investigate how different attributes, such as visual range, distance, speed, opening angles, impact the implementation of perception models in virtual simulations.
5. Collaborate in a multidisciplinary team to integrate perception and environment modeling techniques.
6. Communicate technical results effectively through reports and presentations.

STUDENT BACKGROUND REQUIREMENTS:

Required:

- Programming: Proficiency in Python or C++ with experience in data processing libraries (e.g., NumPy, OpenCV, PCL). *At least 2 students need to have prior knowledge in Python or C++.*
- Data Analysis: Familiarity with signal processing, sensor data characteristics.
- Basic understanding of coordinate systems and geometric transformations.

Preferable (Good to Have):

- Experience with sensor fusion techniques in autonomous driving or robotics.
- Familiarity with machine learning or statistical modeling for sensor data interpretation.

SUPERVISORS

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STAKEHOLDERS

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